

Spring 2022

# Applied quantitative Methods for Non-quantitative doctoral researchers

## Session 06: Correlation analysis

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# Analysis objectives

1. Describing data with regards to the variables of interest

2. Looking for differences between groups

3. Looking for relationships between different variables

4. Assessing the influence of changes in independent variables on dependent variables

# Testing relationships

## Nominal



By Anne R. Carey and Chad Palmer, USA TODAY  
Source: hudson-index.com

- ☒ Chi square
- ☐ Proportions test

## Ordinal



- ☒ Chi square
- ☒ Rank correlation (Kendall'  $\tau$ )
- ☒ Spearman correlation

## Interval



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- ☒ Pearson correlation
- ☒ Spearman correlation
- ☒ Bivariate regression

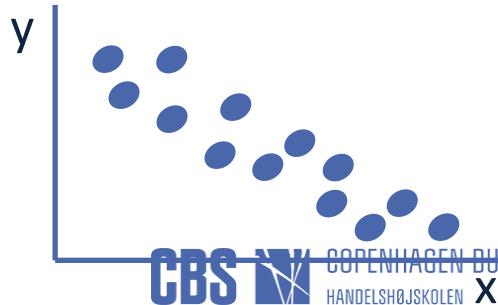
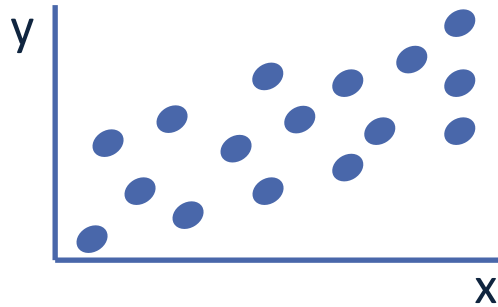
## Ratio



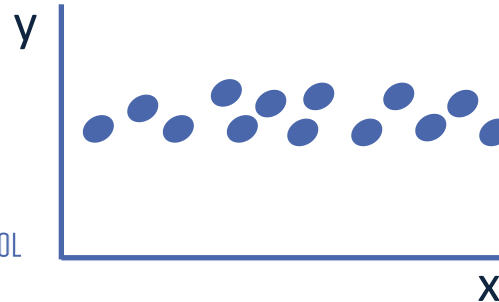
- ☒ Pearson correlation
- ☒ Spearman correlation
- ☒ Bivariate regression

# Scatter plots

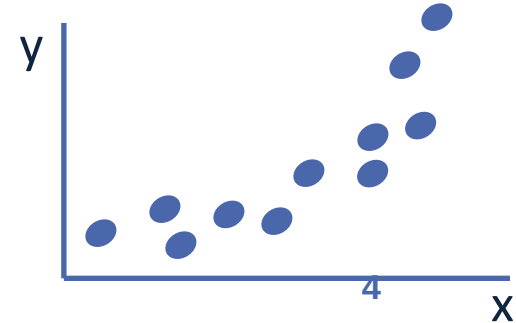
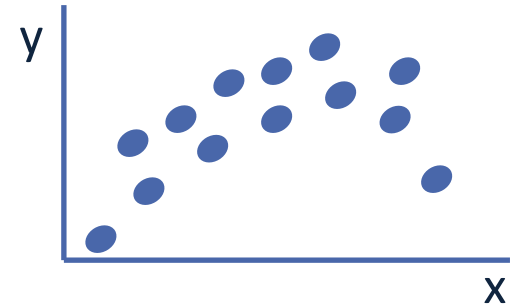
Linear relationships



No relationship



Curvilinear relationships



# Covariance

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We learned the measure of **variance**, explaining how one variable varies (around its own mean)

Sample Variance

$$s^2 = \frac{\sum (x - \bar{x})^2}{n - 1}$$

Sample Standard Deviation

$$s = \sqrt{\frac{\sum (x - \bar{x})^2}{n - 1}}$$

The **covariance** is a measure explaining how two variables vary together (around their own means)

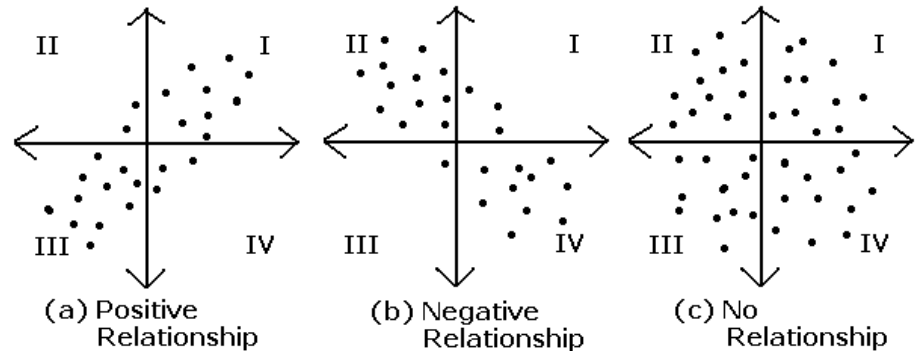
$$COV(X, Y) = \frac{\sum_{i=1}^n (X_i - \bar{X}) (Y_i - \bar{Y})}{n - 1}$$

# The correlation coefficient

- Covariance is dependent on the scale of measurement, so we have to standardize our variables.
- The standardized covariance is Pearson' correlation!
- We use the Standard Deviations of X and Y.

$$S_y = \sqrt{\frac{1}{N-1} \sum_{i=1}^N (y_i - \bar{y})^2}$$

$$S_x = \sqrt{\frac{1}{N-1} \sum_{i=1}^N (x_i - \bar{x})^2}$$



# The correlation coefficient

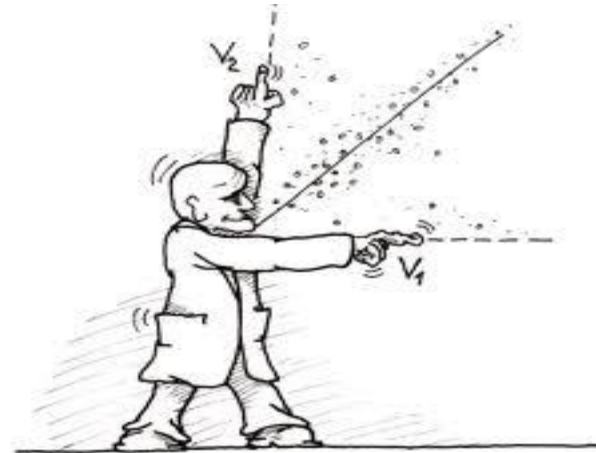
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The correlation is the standardized covariance

$$r = r_{xy} = \frac{\text{Cov}(x, y)}{S_x \times S_y}$$

# The correlation coefficient

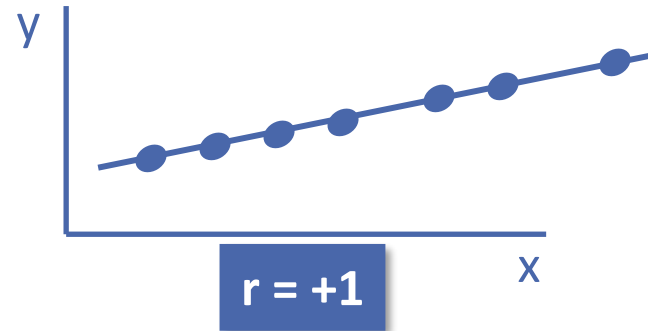
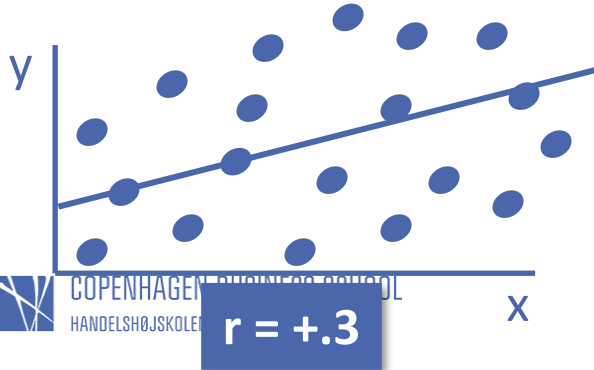
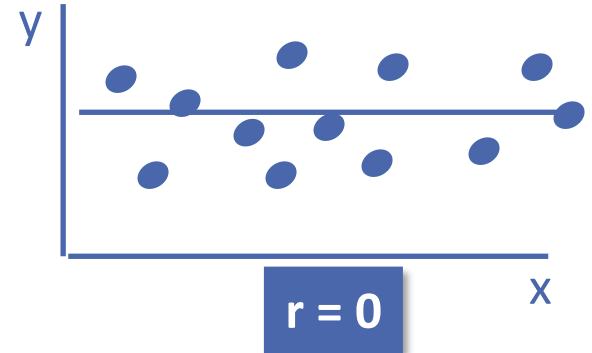
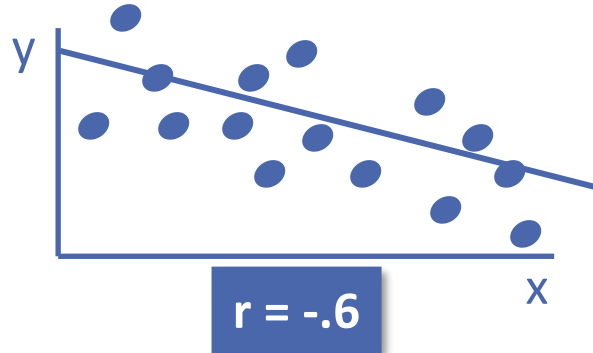
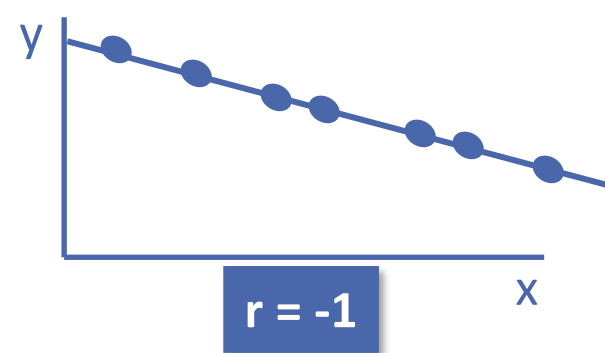
- Correlation quantifies the degree of linear association between two variables
- The correlations coefficient  $r$  measures the strength and direction of the linear relationship
- Unit free
- Value of  $r$  is between -1 and +1
- Positive:  $X \uparrow, Y \uparrow$
- Negative:  $X \uparrow, Y \downarrow$





# The correlation coefficient

Relationship scatter plot & correlation



# Covariance/Correlation in STATA

```
. sum X Y, det
```

| X           |    |          |             |     |
|-------------|----|----------|-------------|-----|
| Percentiles |    | Smallest |             |     |
| 1%          | 8  | 8        |             |     |
| 5%          | 8  | 10       |             |     |
| 10%         | 8  | 12       | Obs         | 3   |
| 15%         | 8  | .        | Sum of Wgt. | 3   |
|             |    |          |             |     |
| 50%         | 10 |          | Mean        | 10  |
|             |    | Largest  | Std. Dev.   | 2   |
| 75%         | 12 | .        |             |     |
| 90%         | 12 | 8        | Variance    | 4   |
| 95%         | 12 | 10       | Skewness    | 0   |
| 99%         | 12 | 12       | Kurtosis    | 1.5 |

| Y           |    |          |             |     |
|-------------|----|----------|-------------|-----|
| Percentiles |    | Smallest |             |     |
| 1%          | 9  | 9        |             |     |
| 5%          | 9  | 10       |             |     |
| 10%         | 9  | 11       | Obs         | 3   |
| 15%         | 9  | .        | Sum of Wgt. | 3   |
|             |    |          |             |     |
| 50%         | 10 |          | Mean        | 10  |
|             |    | Largest  | Std. Dev.   | 1   |
| 75%         | 11 | .        |             |     |
| 90%         | 11 | 9        | Variance    | 1   |
| 95%         | 11 | 10       | Skewness    | 0   |
| 99%         | 11 | 11       | Kurtosis    | 1.5 |

```
. corr X Y , covariance
(obs=3)
```

|   | X | Y |
|---|---|---|
| X | 4 |   |
| Y | 2 | 1 |

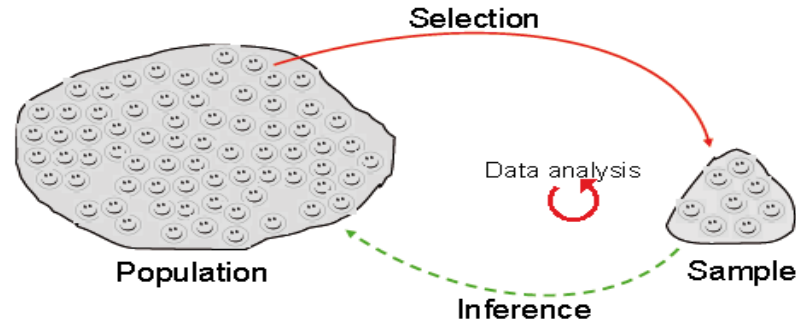
```
. corr X Y
(obs=3)
```

|   | X      | Y      |
|---|--------|--------|
| X | 1.0000 |        |
| Y | 1.0000 | 1.0000 |

| X  | Y  |
|----|----|
| 12 | 11 |
| 10 | 10 |
| 8  | 9  |

# The correlation coefficient

The population correlation coefficient  $\rho$  (rho) measures the strength of the association between the variables



The sample correlation coefficient  $r$  is an estimate of  $\rho$  and is used to measure the strength and direction of the linear relationship in the sample observations

# How to start

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A scatter plot is used to show the relationship between two variables

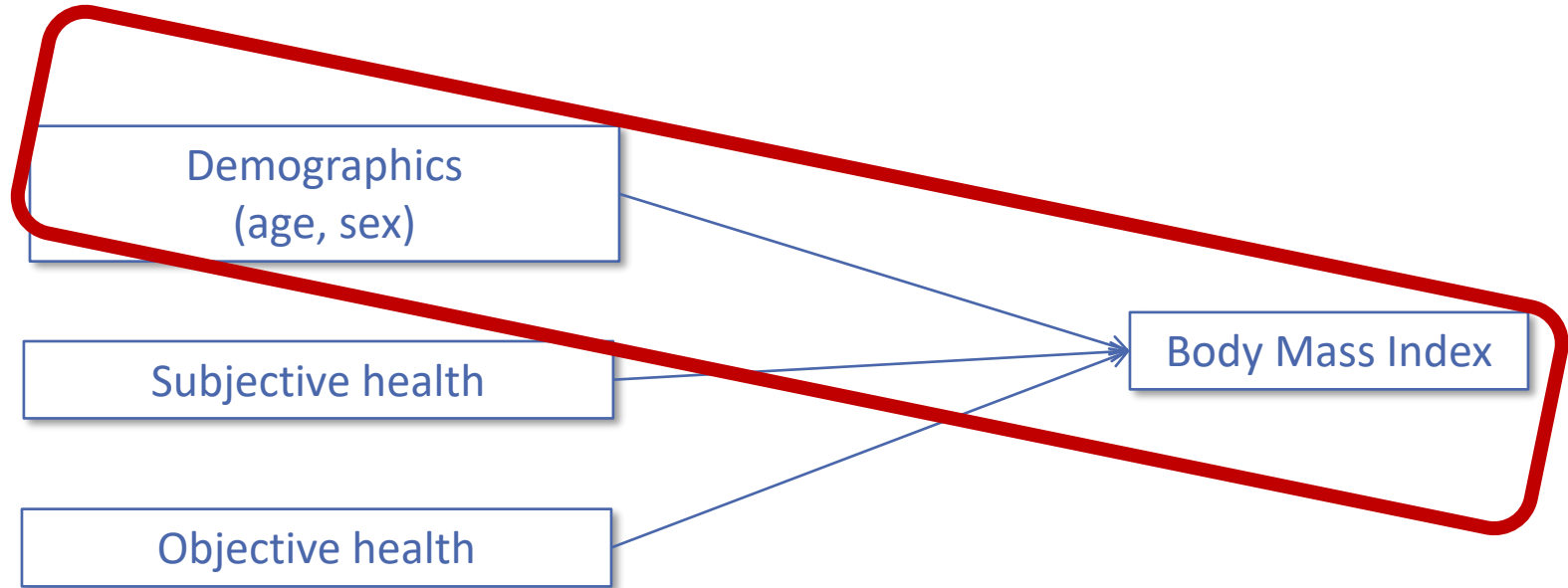
Correlation analysis is used to measure strength and direction of the association (linear relationship) between two variables

Only concerned with strength and direction of the relationship

No causal effect is implied

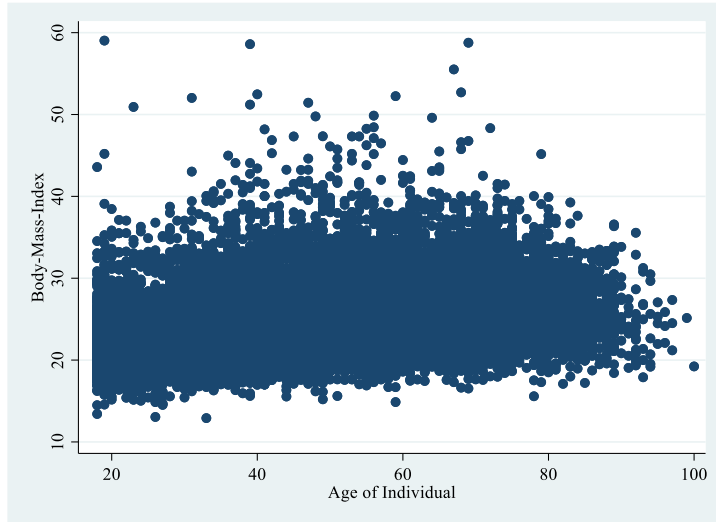
# An example

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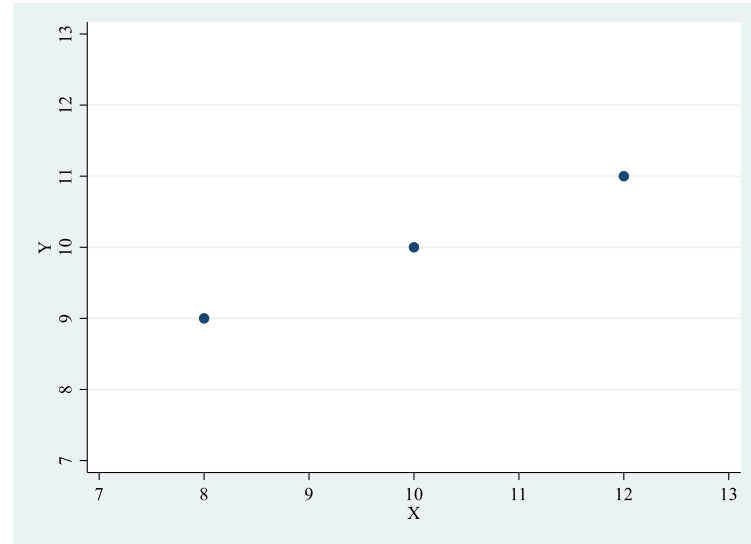


# An example

STATA: scatter h\_bmi age

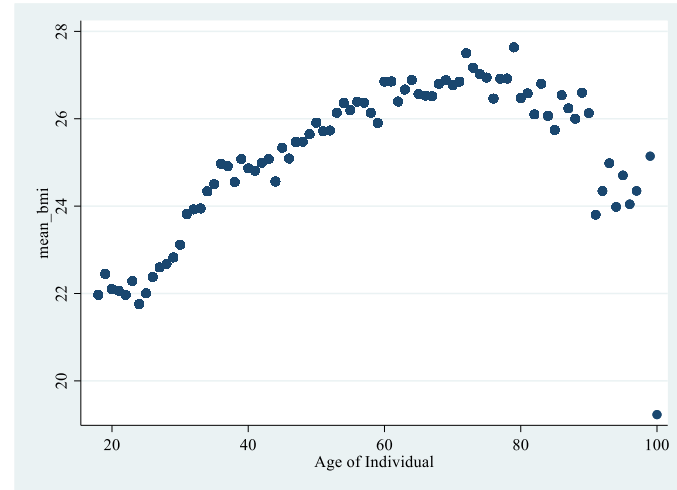
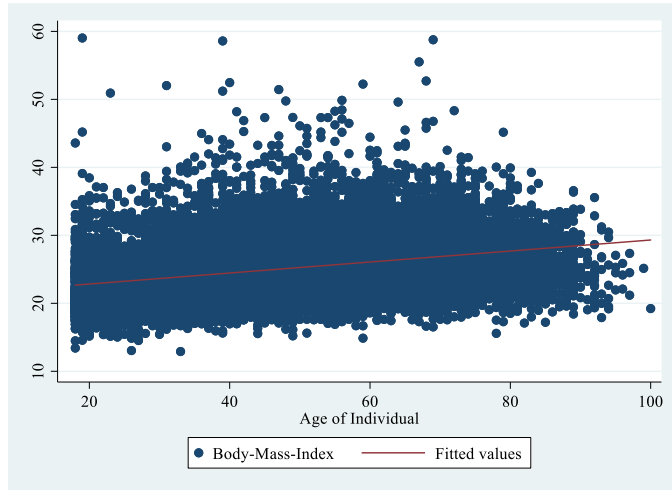


Simplified example: scatter Y X



# An example

We hypothesize that  
... age is related to BMI



# An example

Pearson correlation coefficient  $r$ :

|       | h_bmi  | age    |
|-------|--------|--------|
| h_bmi | 1.0000 |        |
| age   | 0.3227 | 1.0000 |

Correlation between  
age and BMI, but...



# An example

... we want to know if this correlation also exists in the population

➔ There are 6 steps for hypothesis testing?

$H_0: \rho = 0$  (No correlation)

$H_1: \rho \neq 0$  (Correlation exists)

Step 1

Significance level:  $\alpha = .05$

Step 2

$$t_{\text{calc}} = \frac{r}{\sqrt{\frac{1-r^2}{n-2}}} = \frac{0.3227}{\sqrt{\frac{1-(0.3227)^2}{18,122-2}}} = 45.89$$

Step 3

# Things STATA does for you:

If done by hand: After calculating the t-statistic you have look up the critical value in a table:

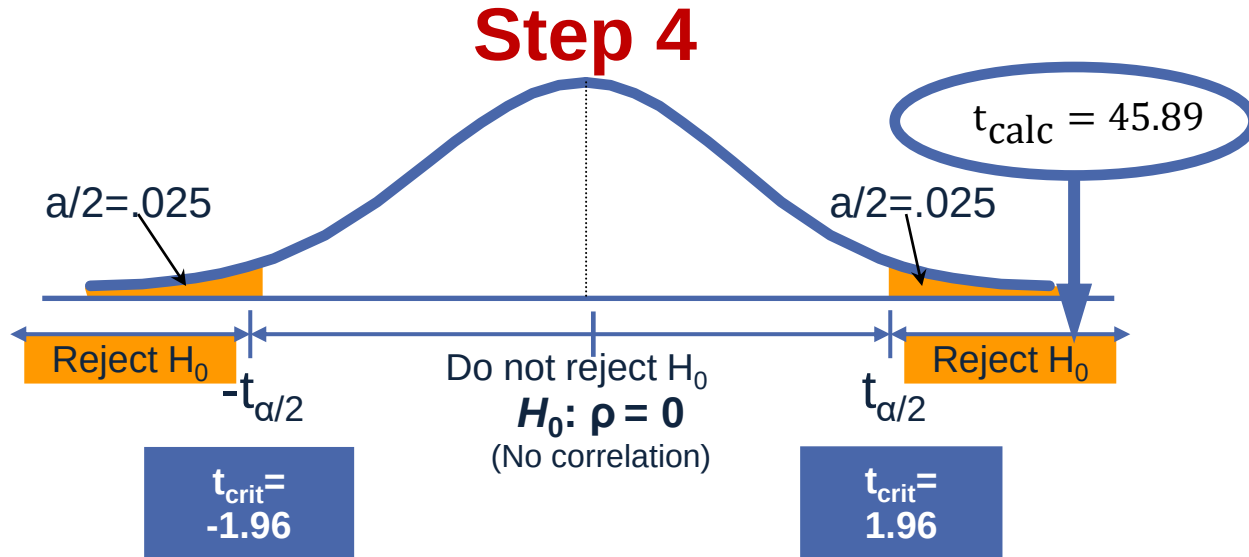
For alpha = .05 the value is 1.960

t Table

| cum. prob<br>one-tail | $t_{.50}$ | $t_{.25}$ | $t_{.20}$ | $t_{.15}$ | $t_{.10}$ | $t_{.05}$ | $t_{.025}$ | $t_{.01}$ | $t_{.005}$ | $t_{.001}$ | $t_{.0005}$ |
|-----------------------|-----------|-----------|-----------|-----------|-----------|-----------|------------|-----------|------------|------------|-------------|
| two-tails             | 1.00      | 0.50      | 0.40      | 0.30      | 0.20      | 0.10      | 0.05       | 0.02      | 0.01       | 0.002      | 0.001       |
| df                    |           |           |           |           |           |           |            |           |            |            |             |
| 1                     | 0.000     | 1.000     | 1.376     | 1.963     | 3.078     | 6.314     | 12.71      | 31.82     | 63.66      | 318.31     | 636.62      |
| 2                     | 0.000     | 0.816     | 1.061     | 1.386     | 1.886     | 2.920     | 4.303      | 6.965     | 9.925      | 22.327     | 31.599      |
| 3                     | 0.000     | 0.765     | 0.978     | 1.250     | 1.638     | 2.353     | 3.182      | 4.541     | 5.841      | 10.215     | 12.924      |
| 4                     | 0.000     | 0.741     | 0.941     | 1.190     | 1.533     | 2.132     | 2.776      | 3.747     | 4.604      | 7.173      | 8.610       |
| 5                     | 0.000     | 0.727     | 0.920     | 1.156     | 1.476     | 2.015     | 2.571      | 3.365     | 4.032      | 5.893      | 6.869       |
| 6                     | 0.000     | 0.718     | 0.906     | 1.134     | 1.440     | 1.943     | 2.447      | 3.143     | 3.707      | 5.208      | 5.959       |
| 7                     | 0.000     | 0.711     | 0.896     | 1.119     | 1.415     | 1.895     | 2.365      | 2.998     | 3.499      | 4.785      | 5.408       |
| 8                     | 0.000     | 0.706     | 0.889     | 1.108     | 1.397     | 1.860     | 2.306      | 2.896     | 3.355      | 4.501      | 5.041       |
| 9                     | 0.000     | 0.703     | 0.883     | 1.100     | 1.383     | 1.833     | 2.262      | 2.821     | 3.250      | 4.297      | 4.781       |
| 10                    | 0.000     | 0.700     | 0.879     | 1.093     | 1.372     | 1.812     | 2.228      | 2.764     | 3.169      | 4.144      | 4.587       |
| 11                    | 0.000     | 0.697     | 0.876     | 1.088     | 1.363     | 1.796     | 2.201      | 2.718     | 3.106      | 4.025      | 4.437       |
| 12                    | 0.000     | 0.695     | 0.873     | 1.083     | 1.356     | 1.782     | 2.179      | 2.681     | 3.055      | 3.930      | 4.318       |
| 13                    | 0.000     | 0.694     | 0.870     | 1.079     | 1.350     | 1.771     | 2.160      | 2.650     | 3.012      | 3.852      | 4.221       |
| 14                    | 0.000     | 0.692     | 0.868     | 1.076     | 1.345     | 1.761     | 2.145      | 2.624     | 2.977      | 3.787      | 4.140       |
| 15                    | 0.000     | 0.691     | 0.866     | 1.074     | 1.341     | 1.753     | 2.131      | 2.602     | 2.947      | 3.733      | 4.073       |
| 16                    | 0.000     | 0.690     | 0.865     | 1.071     | 1.337     | 1.746     | 2.120      | 2.583     | 2.921      | 3.686      | 4.015       |
| 17                    | 0.000     | 0.689     | 0.863     | 1.069     | 1.333     | 1.740     | 2.110      | 2.567     | 2.898      | 3.646      | 3.965       |
| 18                    | 0.000     | 0.688     | 0.862     | 1.067     | 1.330     | 1.734     | 2.101      | 2.552     | 2.878      | 3.610      | 3.922       |
| 19                    | 0.000     | 0.688     | 0.861     | 1.066     | 1.328     | 1.729     | 2.093      | 2.539     | 2.861      | 3.579      | 3.883       |
| 20                    | 0.000     | 0.687     | 0.860     | 1.064     | 1.325     | 1.725     | 2.086      | 2.528     | 2.845      | 3.552      | 3.850       |
| 21                    | 0.000     | 0.686     | 0.859     | 1.063     | 1.323     | 1.721     | 2.080      | 2.518     | 2.831      | 3.527      | 3.819       |
| 22                    | 0.000     | 0.686     | 0.858     | 1.061     | 1.321     | 1.717     | 2.074      | 2.508     | 2.819      | 3.505      | 3.792       |
| 23                    | 0.000     | 0.685     | 0.858     | 1.060     | 1.319     | 1.714     | 2.069      | 2.500     | 2.807      | 3.485      | 3.768       |
| 24                    | 0.000     | 0.685     | 0.857     | 1.059     | 1.318     | 1.711     | 2.064      | 2.492     | 2.797      | 3.467      | 3.745       |
| 25                    | 0.000     | 0.684     | 0.856     | 1.058     | 1.316     | 1.708     | 2.060      | 2.485     | 2.787      | 3.450      | 3.725       |
| 26                    | 0.000     | 0.684     | 0.856     | 1.058     | 1.315     | 1.706     | 2.056      | 2.479     | 2.779      | 3.435      | 3.707       |
| 27                    | 0.000     | 0.684     | 0.855     | 1.057     | 1.314     | 1.703     | 2.052      | 2.473     | 2.771      | 3.421      | 3.690       |
| 28                    | 0.000     | 0.683     | 0.855     | 1.056     | 1.313     | 1.701     | 2.048      | 2.467     | 2.763      | 3.408      | 3.674       |
| 29                    | 0.000     | 0.683     | 0.854     | 1.055     | 1.311     | 1.699     | 2.045      | 2.462     | 2.756      | 3.396      | 3.659       |
| 30                    | 0.000     | 0.683     | 0.854     | 1.055     | 1.310     | 1.697     | 2.042      | 2.457     | 2.750      | 3.385      | 3.646       |
| 40                    | 0.000     | 0.681     | 0.851     | 1.050     | 1.303     | 1.684     | 2.021      | 2.423     | 2.704      | 3.307      | 3.551       |
| 60                    | 0.000     | 0.679     | 0.848     | 1.045     | 1.296     | 1.671     | 2.000      | 2.390     | 2.660      | 3.232      | 3.480       |
| 80                    | 0.000     | 0.678     | 0.846     | 1.043     | 1.292     | 1.664     | 1.990      | 2.374     | 2.639      | 3.195      | 3.416       |
| 100                   | 0.000     | 0.677     | 0.845     | 1.042     | 1.290     | 1.660     | 1.984      | 2.364     | 2.626      | 3.174      | 3.390       |
| 1000                  | 0.000     | 0.675     | 0.842     | 1.037     | 1.282     | 1.646     | 1.962      | 2.330     | 2.581      | 3.098      | 3.300       |
| <b>Z</b>              | 0.000     | 0.674     | 0.842     | 1.036     | 1.282     | 1.645     | 1.960      | 2.326     | 2.576      | 3.090      | 3.291       |
|                       | 0%        | 50%       | 60%       | 70%       | 80%       | 90%       | 95%        | 98%       | 99%        | 99.8%      | 99.9%       |
| Confidence Level      |           |           |           |           |           |           |            |           |            |            |             |

| <b>Z</b> | 0.000 | 0.674 | 0.842 | 1.036 | 1.282 | 1.645 | 1.960 | 2.326 | 2.576 | 3.090 | 3.291 |
|----------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|
|          | 0%    | 50%   | 60%   | 70%   | 80%   | 90%   | 95%   | 98%   | 99%   | 99.8% | 99.9% |

# An example



**Step 5**

Decision:  
Reject  $H_0$

**And  
step 6?**

**Conclusion:**

There is **evidence** for a (*against no*) linear relationship between age and BMI

# An example

```
. pwcorr h_bmi age sex hhincnet satlife hleis_wd hjob_wd , o star(.05)
```

|          | h_bmi             | age               | sex               | hhincnet          | satlife          | hleis_wd          | hjob_wd         |
|----------|-------------------|-------------------|-------------------|-------------------|------------------|-------------------|-----------------|
| h_bmi    | 1.0000<br>18122   |                   |                   |                   |                  |                   |                 |
| age      | 0.3227*<br>18122  | 1.0000<br>20578   |                   |                   |                  |                   |                 |
| sex      | -0.1468*<br>18122 | 0.0113<br>20578   | 1.0000<br>20578   |                   |                  |                   |                 |
| hhincnet | -0.0912*<br>18122 | -0.1023*<br>20578 | -0.0565*<br>20578 | 1.0000<br>20578   |                  |                   |                 |
| satlife  | -0.0892*<br>18066 | -0.0493*<br>20519 | 0.0110<br>20519   | 0.1618*<br>20519  | 1.0000<br>20519  |                   |                 |
| hleis_wd | 0.0515*<br>17494  | 0.1638*<br>19937  | -0.0290*<br>19937 | -0.0828*<br>19937 | 0.0050<br>19891  | 1.0000<br>19937   |                 |
| hjob_wd  | -0.0893*<br>17082 | -0.4225*<br>19467 | -0.2165*<br>19467 | 0.2243*<br>19467  | 0.0577*<br>19419 | -0.3457*<br>19068 | 1.0000<br>19467 |



## Interval/Ratio vs. Ordinal scale

Correlation between X & Y

X => Interval/Ratio

Y => Interval/Ratio

**Pearson** Correlation

Correlation between X & Y

At least one variable ordinal

X => Ordinal

Y => Ordinal

**Spearman** Correlation

# An example

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## Correlation does not mean causation!!!

- Correlation is (almost always) a necessary but not sufficient condition for causation:
- By a lurking third variable  
(e.g., shoe size and literacy for children)
- Coincidental relationship?  
(e.g., number of storks and birth rate)



# Exercise from last year

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We collected the following data:

- Hand span
- Height
- Gender

Let's look at the data!

